

ORGANIC CHEMISTRY
ASYNCHRONOUS ACTIVITY 3



NAME IT, DRAW IT



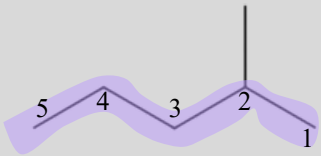
Windel M. Narito
BSChE 2A



PART A - STRUCTURE → NAME

CARD A01 - FOUNDATION • MOLECULAR FORMULA: C_6H_{14}

1. NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the right-hand end.

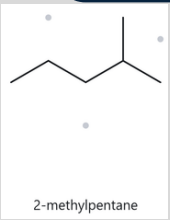
2. NAMING TRAIL

- **Parent**: longest chain = 5 C → pentane.
- **Numbering**: left → right gives 4; right → left gives 2. 2 < 4, so number from the right.
- **Substituents**: methyl on C2 → 2-methyl.

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



- **Returned Name**: 2-methylpentane

✅ Match

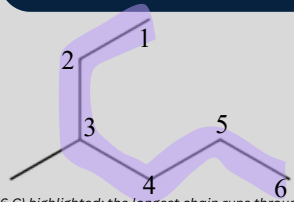
The rebuilt structure is a 5-C chain with one CH3 on the second carbon. The tool applies the same rules as the naming trail: longest chain first, then lowest locant (2 beats 4). Both therefore give the same name.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A02 - FOUNDATION • MOLECULAR FORMULA: C_7H_{16}

1. NUMBERED PARENT



Parent (6 C) highlighted; the longest chain runs through the branch, not along the horizontal.

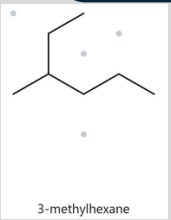
2. NAMING TRAIL

- **Parent**: longest chain = 6 C → hexane. The horizontal path is only 5 C.
- **Numbering**: top → bottom gives 3; the other direction gives 4. 3 < 4, so number from the top.
- **Substituents**: the short left stub is a methyl on C3 → 3-methyl

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



- **Returned Name**: 3-methylhexane

✅ Match

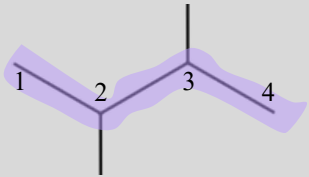
The structure has a 6-C path through the branch point, with one methyl on the third carbon. The tool traces the longest chain through branch points, so hexane is chosen over the 5-C horizontal path, and 3 is the lowest locant.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A03 - FOUNDATION • MOLECULAR FORMULA: C_6H_{14}

1. NUMBERED PARENT



Parent (4 C) highlighted; the two vertical stubs are substituents.

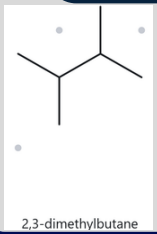
2. NAMING TRAIL

- **Parent**: longest chain = 4 C → butane.
- **Numbering**: both directions give 2,3, so the sets tie and either direction is fine.
- **Substituents**: methyl on C2 and methyl on C3 → 2,3-dimethyl

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



2,3-dimethylbutane

- **Returned Name**: 2,3-dimethylbutane

✅ Match

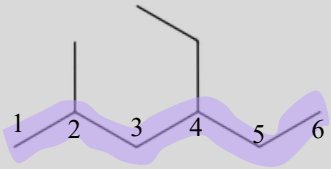
The structure is a 4-C chain with one methyl on each middle carbon. The locant set is {2,3} from both ends, and the tool writes identical groups with the prefix di- and a comma between locants.

✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

PART A - STRUCTURE → NAME

CARD A04 - DEVELOPING • MOLECULAR FORMULA: C_9H_{20}

1. NUMBERED PARENT



Parent (6 C) highlighted; carbons numbered from the left-hand end.

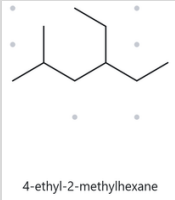
2. NAMING TRAIL

- **Parent**: 6 C → hexane. Another 6-C path runs through the ethyl, but it carries the same two substituents (a methyl and an ethyl), so it leads to the same name.
- **Numbering**: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.
- **Substituents**: ethyl on C4, then methyl on C2 → 4-ethyl-2-methyl. Alphabetical order sets the citation order, not the locants.

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



- **Returned Name**: 4-ethyl-2-methylhexane

✅ Match

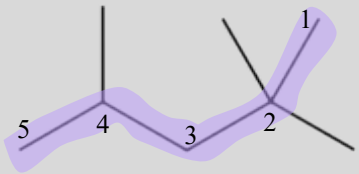
The tool numbers from the end that gives {2,4}, not {3,5}. It cites ethyl before methyl because e comes before m alphabetically, even though ethyl has the higher locant.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A05 - DEVELOPING • MOLECULAR FORMULA: C_8H_{18}

1. NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the right-hand end.

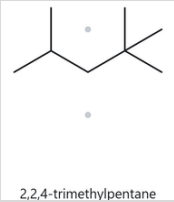
2. NAMING TRAIL

- **Parent**: longest chain = 5 C → pentane.
- **Numbering**: left → right gives 2,4,4; right → left gives 2,2,4. The first terms tie at 2. At the second term 2 < 4, so number from the right.
- **Substituents**:methyl on C2, C2 and C4 → 2,2,4-trimethyl.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



- **Returned Name**: 2,2,4-trimethylpentane

✅ Match

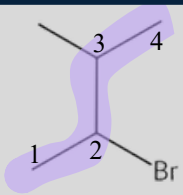
The tool compares locant sets term by term, not by sum. {2,2,4} beats {2,4,4} at the second term, and the locant 2 is repeated once for each methyl on that carbon.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A06 - DEVELOPING • MOLECULAR FORMULA: $C_5H_{11}Br$

1. NUMBERED PARENT



Parent (4 C) highlighted; Br and the other top methyl are substituents.

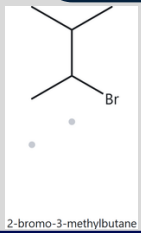
2. NAMING TRAIL

- **Parent**: longest chain = 4 C → butane.
- **Numbering**: both directions give {2,3}, so the locant sets tie.
- **Tie-break**: bromo (b) precedes methyl (m) alphabetically, so bromo gets C2.
- **Substituents**: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



2-bromo-3-methylbutane

- **Returned Name**: 2-bromo-3-methylbutane

✅ Match

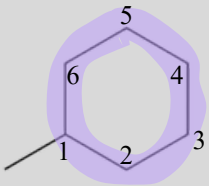
The locants tie at {2,3}, so the tool applies the alphabetical tie-break and gives the lower number to bromo. The halogen is written as a prefix (bromo) and is not part of the parent name.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A07 - DEVELOPING • MOLECULAR FORMULA: C_7H_{14}

1. NUMBERED PARENT



Parent (6-C ring) highlighted; the methyl is a substituent, not part of the parent.

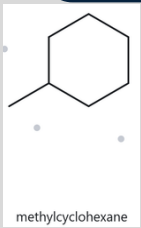
2. NAMING TRAIL

- **Parent**: the ring is the parent, 6 C → cyclohexane.
- **Numbering**: there is only one substituent, so there are no locants to compare. The substituent carbon is C1, and no locant is written.
- **Substituents**: methyl.

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



- **Returned Name**: methylcyclohexane

✅ Match

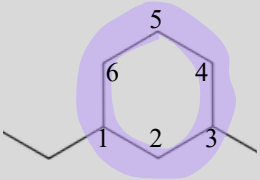
The structure is a 6-C ring with one CH3. The tool takes the ring as the parent and, because there is only one substituent, writes no locant (the name is not 1-methylcyclohexane).

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

PART A - STRUCTURE → NAME

CARD A08 - CHALLENGE • MOLECULAR FORMULA: C_9H_{18}

1. NUMBERED PARENT



Parent (6-C ring) highlighted; numbered the short way round from the ethyl carbon.

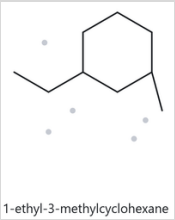
2. NAMING TRAIL

- **Parent**: the ring is the parent → cyclohexane.
- **Numbering**: going the short way round gives {1,3} whichever substituent is C1. The long way round gives {1,5}. {1,3} wins, and the two short-way sets tie.
- **Tie-break**: ethyl (e) precedes methyl (m), so ethyl gets C1.
- **Substituents**: ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK


1-ethyl-3-methylcyclohexane

- **Returned Name**: 1-ethyl-3-methylcyclohexane

✅ Match

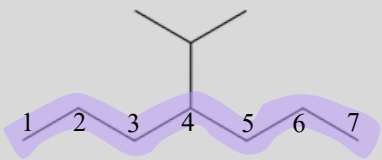
The two substituents are separated by one ring carbon, so the lowest locant set is {1,3}. The sets tie in both directions, so the alphabetical rule places ethyl on C1.

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

PART A - STRUCTURE → NAME

CARD A09 - CHALLENGE • MOLECULAR FORMULA: $C_{10}H_{22}$

1. NUMBERED PARENT



Parent (6-C ring) highlighted; numbered the short way round from the ethyl carbon.

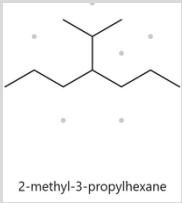
2. NAMING TRAIL

- **Parent**: Parent (7 C) highlighted; the 3-C branch is a substituent.
- **Numbering**: both directions give 4, so the sets tie.
- **Substituents**: a 3-carbon branch attached through its middle carbon is isopropyl, also written (1-methylethyl) or propan-2-yl, on C4.

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



2-methyl-3-propylhexane

- **Returned Name**: 2-methyl-3-propylhexane

❌Did Not Match

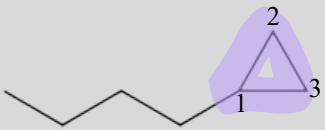
The tool chose a 6-C parent, but a 7-C chain exists through the two propyl arms. The longest-chain rule requires heptane, so the tool's name breaks the rule and the trail's name is kept.

✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

PART A - STRUCTURE → NAME

CARD A10 - CHALLENGE • MOLECULAR FORMULA: C_7H_{14}

1. NUMBERED PARENT



Parent (3-C ring) highlighted; the 4-C chain is a substituent.

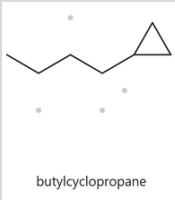
2. NAMING TRAIL

- **Parent**: the ring is the parent under this activity's rule, even though the chain (4 C) is longer than the ring (3 C) → cyclopropane.
- **Numbering**: one substituent, so no locant is needed.
- **Substituents**: unbranched 4-carbon chain → butyl.

3. MY NAME

butylcyclopropane

4. THE ALKANE CHECK



- **Returned Name**: butylcyclopropane

✅ Match

The tool follows the same ring-as-parent rule as this activity. It names the 3-C ring as the parent and the chain as butyl, with no locant for a single substituent.

✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

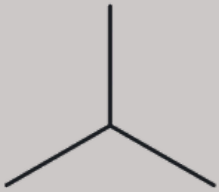
PART B - NAME → STRUCTURE

CARD B01 - FOUNDATION • **Given Name: 2-methylpropane**

1. NAME DECODED

- **Parent:** propane → a 3-carbon chain.
- **Substituents:** methyl on C2.
- **Expected formula:** 3 + 1 = 4 C; open chain → C_nH_{2n+2} = C_4H_{10}

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_4H_{10}
- **Open chain:** C_nH_{2n+2} = C_4H_{10} ✓

4. BACK-CHECK

- **Returned by The Alkane:** 2-methylpropane
- ✓ **Same as the given name**
- The longest chain is 3 C (propane) with one methyl on the middle carbon, so the locant is 2 from either end.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

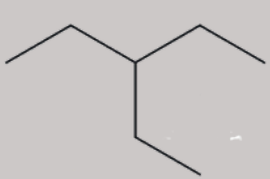
PART B - NAME → STRUCTURE

CARD B02 - FOUNDATION • **Given Name: 3-ethylpentane**

1. NAME DECODED

- **Parent:** pentane (5-carbon chain)
- **Substituents:** ethyl on C3.
- **Expected formula:** 5 + 2 = 7 C; open chain → C_7H_{16}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_7H_{16}
- **Open chain:** C_nH_{2n+2} = C_7H_{16} ✓

4. BACK-CHECK

- **Returned by The Alkane:** 2-3-ethylpentane
- ✓ **Same as the given name**
- The longest chain is 5 C (pentane) with an ethyl on the middle carbon, so the locant is 3 from either end.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

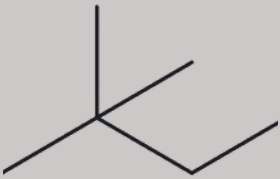
PART B - NAME → STRUCTURE

CARD B03 - FOUNDATION • **Given Name: 2,2-dimethylbutane**

1. NAME DECODED

- **Parent:** butane (4-carbon chain)
- **Substituents:** two methyls, both on C2.
- **Expected formula:** 4 + 2 = 6 C; open chain → C_6H_{14}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_6H_{14}.
- **Open chain:** $C_nH_{2n+2} = C_6H_{14}$ ✓

4. BACK-CHECK

- **Returned by The Alkane:** 2,2-dimethylbutane
- ✓ **Same as the given name**
- The longest chain is 4 C (butane). Numbering gives 2,2 from one end and 3,3 from the other, so 2,2 wins.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

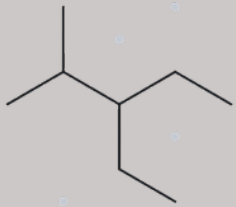
CARD B04 - DEVELOPING

Given Name: 3-ethyl-2-methylpentane

1. NAME DECODED

- **Parent:** pentane (5-carbon chain)
- **Substituents:** methyl on C2; ethyl on C3. Ethyl is cited first because e comes before m, even though its locant is higher.
- **Expected formula:** 5 + 1 + 2 = 8 C; open chain → C_8H_{18}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_8H_{18}.
- **Open chain:** C_nH_{2n+2} = C_8H_{18} ✓

4. BACK-CHECK

- **Returned by The Alkane:** 3-ethyl-2-methylpentane
- ✓ **Same as the given name**
- Numbering gives {2,3} from one end and {3,4} from the other, so {2,3} wins. Ethyl is cited before methyl alphabetically.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

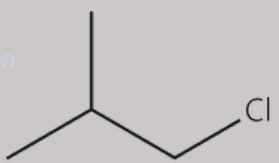
CARD B05 - DEVELOPING

Given Name: 1-chloro-2-methylpropane

1. NAME DECODED

- **Parent:** propane (3-carbon chain)
- **Substituents:** chloro on C1; methyl on C2.
- **Expected formula:** 5 + 1 + 2 = 8 C; open chain → C_8H_{18}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_4H_9Cl.
- **Open chain:** C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓

4. BACK-CHECK

- **Returned by The Alkane:** 1-chloro-2-methylpropane
- ✓ **Same as the given name**
- Numbering gives {1,2} from the Cl end and {2,3} from the other, so {1,2} wins. Chloro is cited before methyl alphabetically.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

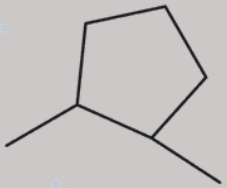
PART B - NAME → STRUCTURE

CARD B06 - DEVELOPING • **Given Name: 1,2-dimethylcyclopentane**

1. NAME DECODED

- **Parent:** cyclopentane (5-carbon ring)
- **Substituents:** methyl on C1; methyl on C2.
- **Expected formula:** 5 + 2 = 7 C; one ring → C_7H_{14}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_7H_{14}.
- **One ring:** C_nH_{2n} = C_7H_{14} ✓

4. BACK-CHECK

- **Returned by The Alkane:** 1,2-dimethylcyclopentane
- ✓ **Same as the given name**
- The ring is the parent. Numbering the short way round gives {1,2}, and the long way gives {1,5}, so {1,2} wins.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

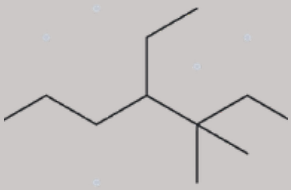
CARD B07 - DEVELOPING

Given Name: 4-ethyl-3,3-dimethylheptane

1. NAME DECODED

- **Parent:** heptane (7-carbon chain)
- **Substituents:** two methyls on C3 (3,3-dimethyl); ethyl on C4. Ethyl is cited first because e comes before m.
- **Expected formula:** 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** $C_{11}H_{24}$.
- **Open chain:** C_nH_{2n+2} = $C_{11}H_{24}$ ✓

4. BACK-CHECK

- **Returned by The Alkane:** 4-ethyl-3,3-dimethylheptane
- ✓ **Same as the given name**
- The longest chain is 7 C. Numbering gives {3,3,4} from one end and {4,5,5} from the other, so {3,3,4} wins. Ethyl is cited before dimethyl.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

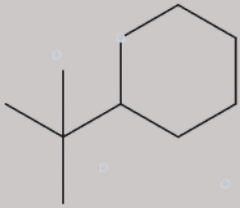
CARD B08 - CHALLENGE

Given Name: tert-butylcyclohexane

1. NAME DECODED

- **Parent:** cyclohexane (6-carbon ring)
- **Substituents:** tert-butyl, $(CH_3)_3C–$, a 4-carbon group, on C1. A single substituent on a ring needs no locant.
- **Expected formula:** 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** $C_{10}H_{20}$.
- **One ring:** $C_nH_{2n} = C_{10}H_{20}$ ✓

4. BACK-CHECK

- **Returned by The Alkane:** (1,1-dimethylethyl)cyclohexane
- ✓ **Same as the given name**
- tert-Butyl and (1,1-dimethylethyl) are equivalent names for the same group (Appendix B.3).

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

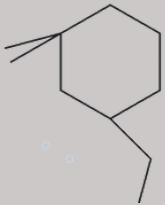
CARD B09 - CHALLENGE

Given Name: 3-ethyl-1,1-dimethylcyclohexane

1. NAME DECODED

- **Parent:** cyclohexane (6-carbon ring)
- **Substituents:** two methyls on C1 (1,1-dimethyl); ethyl on C3.
- **Expected formula:** 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** $C_{10}H_{20}$.
- **One ring:** $C_nH_{2n} = C_{10}H_{20}$ ✓

4. BACK-CHECK

- **Returned by The Alkane:** 3-ethyl-1,1-dimethylcyclohexane
- ✓ **Same as the given name**
- Numbering gives {1,1,3} one way round the ring and {1,1,5} the other, so {1,1,3} wins. Ethyl is cited before dimethyl.

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

PART B - NAME → STRUCTURE

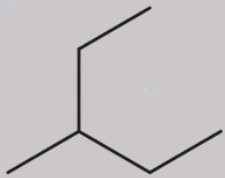
CARD B09 - CHALLENGE

Given Name: 2-ethylbutane

1. NAME DECODED

- **Parent:** butane (4-carbon chain)
- **Substituents:** ethyl on C2.
- **Expected formula:** 4 + 2 = 6 C; open chain → C_6H_{14}.

2. BOND LINE STRUCTURE



3. FORMULA

- **Counted from the drawing:** C_6H_{14}.
- **Open chain:** C_nH_{2n+2} = C_6H_{14} ✓

4. BACK-CHECK

- **Returned by The Alkane:** 3-methylpentane
- ✘ **Different from the given name**
- **Rule broken**: the longest-chain rule (Appendix B.1, step 1). The path ethyl → C2 → C3 → C4 has 5 carbons, which is longer than the 4-carbon "butane" chain. The parent is therefore pentane, and the leftover CH_3 is a methyl on C3 (position 3 from either end).

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded